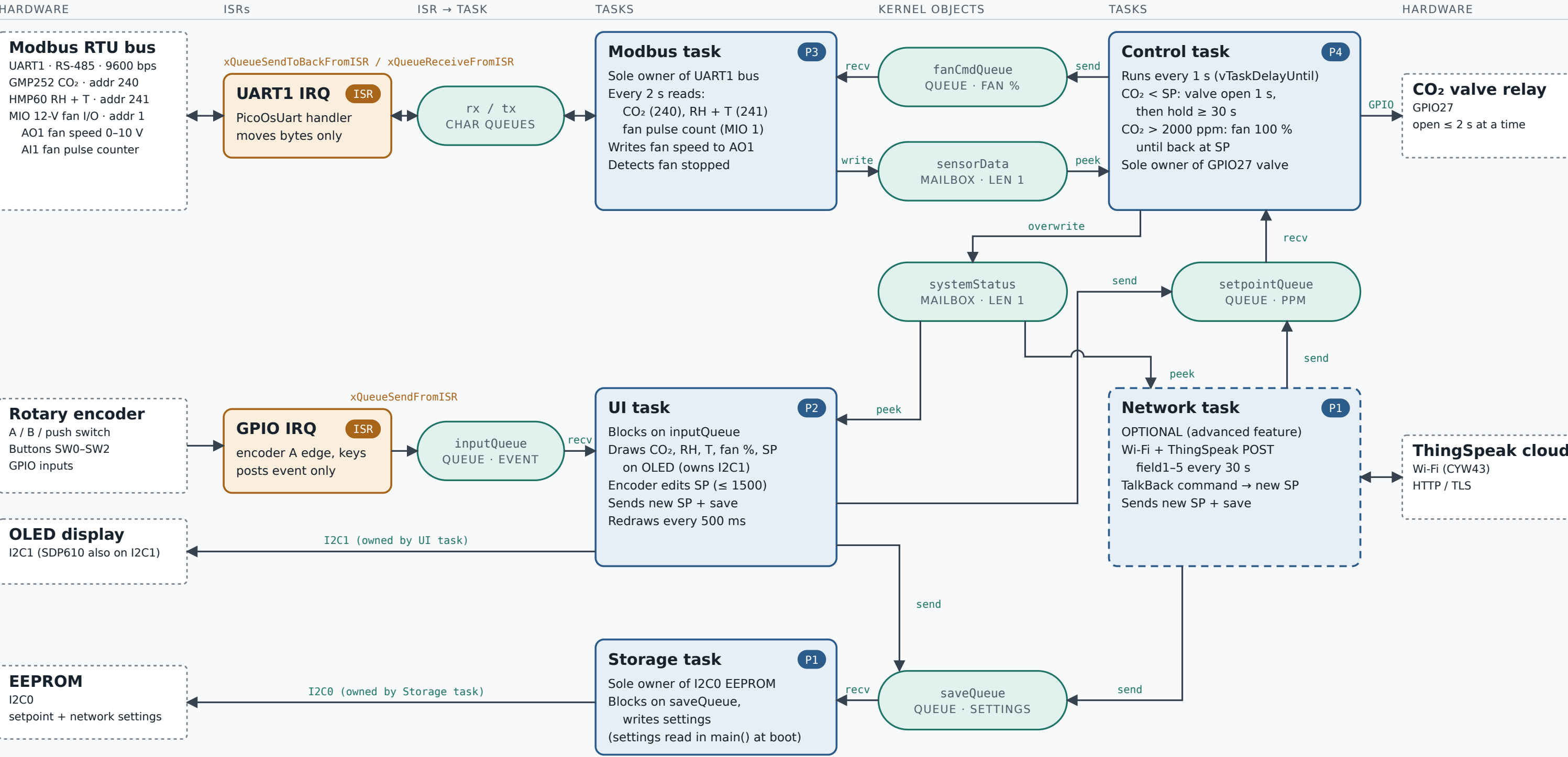


Greenhouse CO₂ controller — FreeRTOS system diagram

RP2040 · 4 tasks + 1 optional · 2 ISRs · every peripheral has exactly one owner task, so no mutex is needed



Task (P = priority, 4 highest) Optional task Interrupt service routine FreeRTOS queue / mailbox Hardware

Mailbox = queue of length 1 written with xQueueOverwrite and read with xQueuePeek, so readers always see the latest value and never block the writer.